

IIT BOMBAY
CHEMICAL ENGINEERING

CL420: Introduction to Biochemical Engineering

Week 2

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The Kingdom of Life

Carl Woese's 1990 rRNA-based scheme: domains differ in cell structure, membrane lipids, and RNA polymerase — not just habitat.

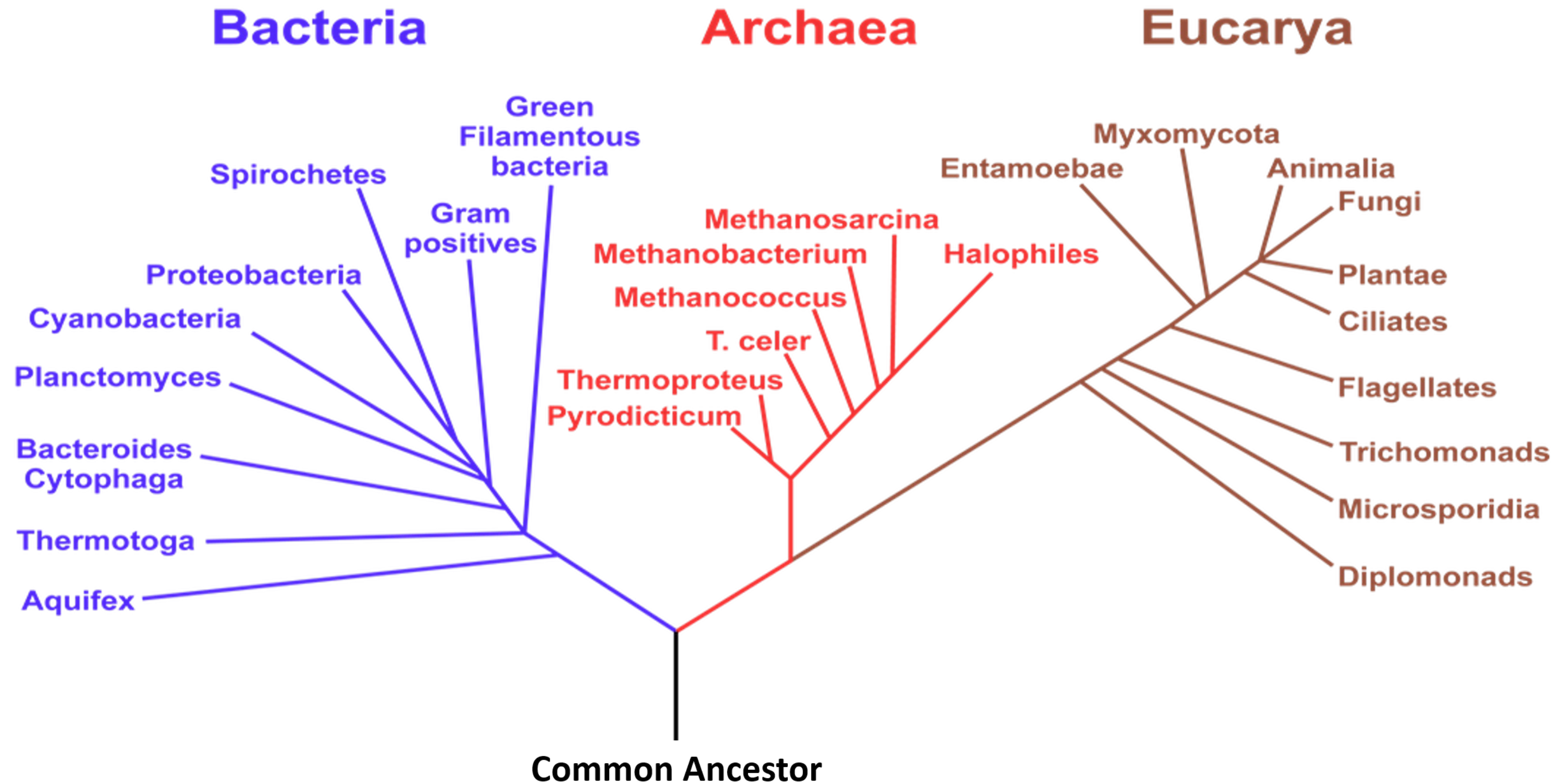


Fig: Classification of the organism into 3 kingdoms of Bacteria, Archaea and Eukarya

Prokaryotes vs Eukaryotes Comparisons

Representative prokaryotic (A) and eukaryotic (B) cell architectures; the structural basis for the property differences is detailed on the next slide

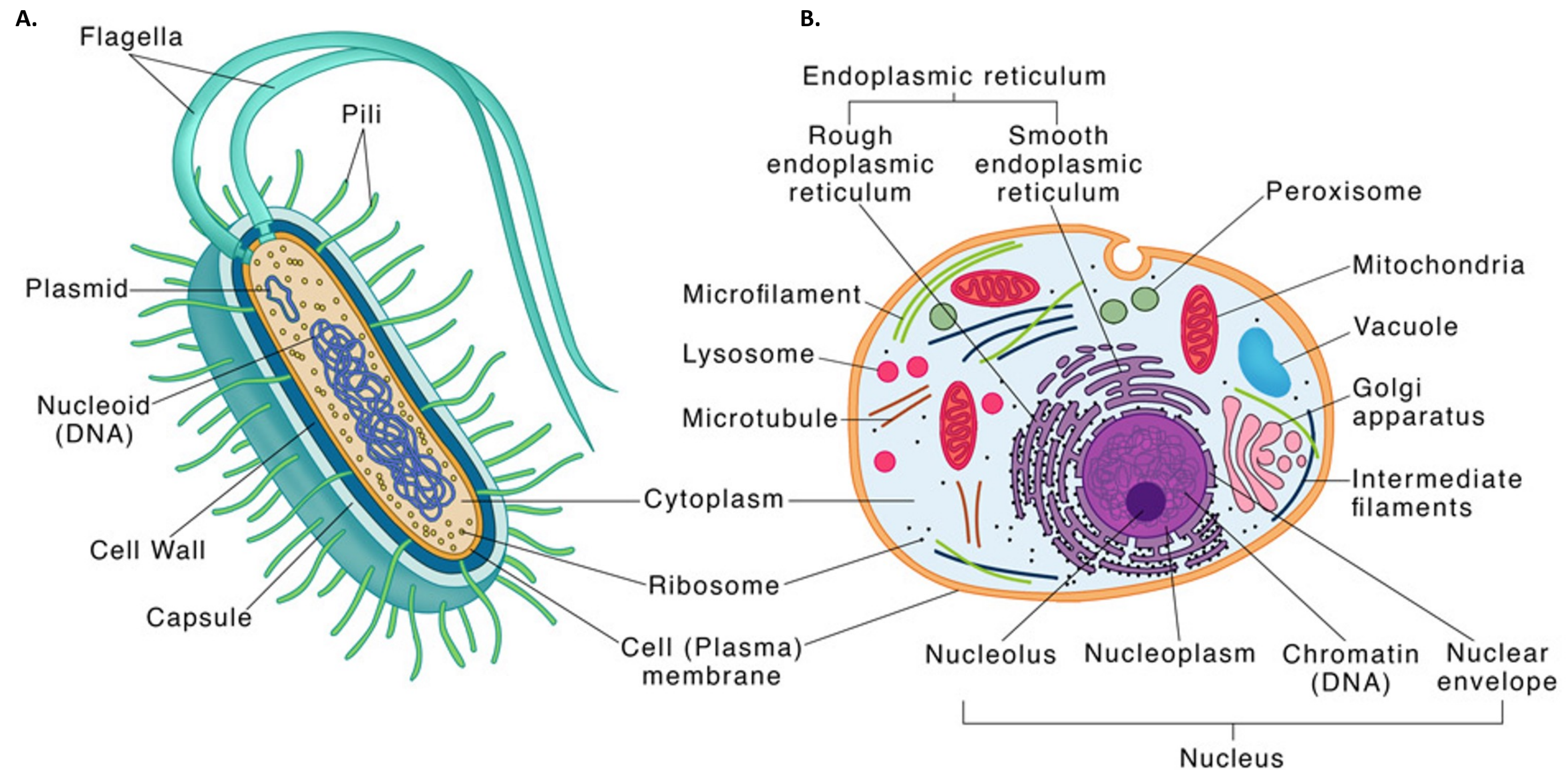


Fig: A.) figure of a prokaryotic organism, B.) figure of an eukaryotic organism

Prokaryotes vs. Eukaryotes Differences

Feature	Prokaryotes	Eukaryotes
Nucleus	None -- DNA sits in an ill-defined "nuclear zone"	Membrane-enclosed nucleus (the defining feature)
Typical size	0.5–3 μm	1,000–10,000x larger by volume
Internal structure	Simple; few or no internal membrane-bound compartments	Complex; distinct membrane-bound compartments (ER, etc.)
Ribosomes	Smaller, free in cytoplasm	Larger, often on endoplasmic reticulum surface
PTM	Absent	Present
Growth rate	Fast -- some double in ~20 min	Slower, more complex cell cycles
Example	<i>Bacillus subtilis</i> , <i>E. coli</i>	<i>Saccharomyces cerevisiae</i> (yeast), <i>P. pastoris</i>

| Lower eukaryotes such as yeast share many similarities with prokaryotes, including high and fast growth rates and minimal growth requirements.

If both prokaryotes and lower eukaryotes can produce proteins, why are cell lines like CHO or HEK necessary?

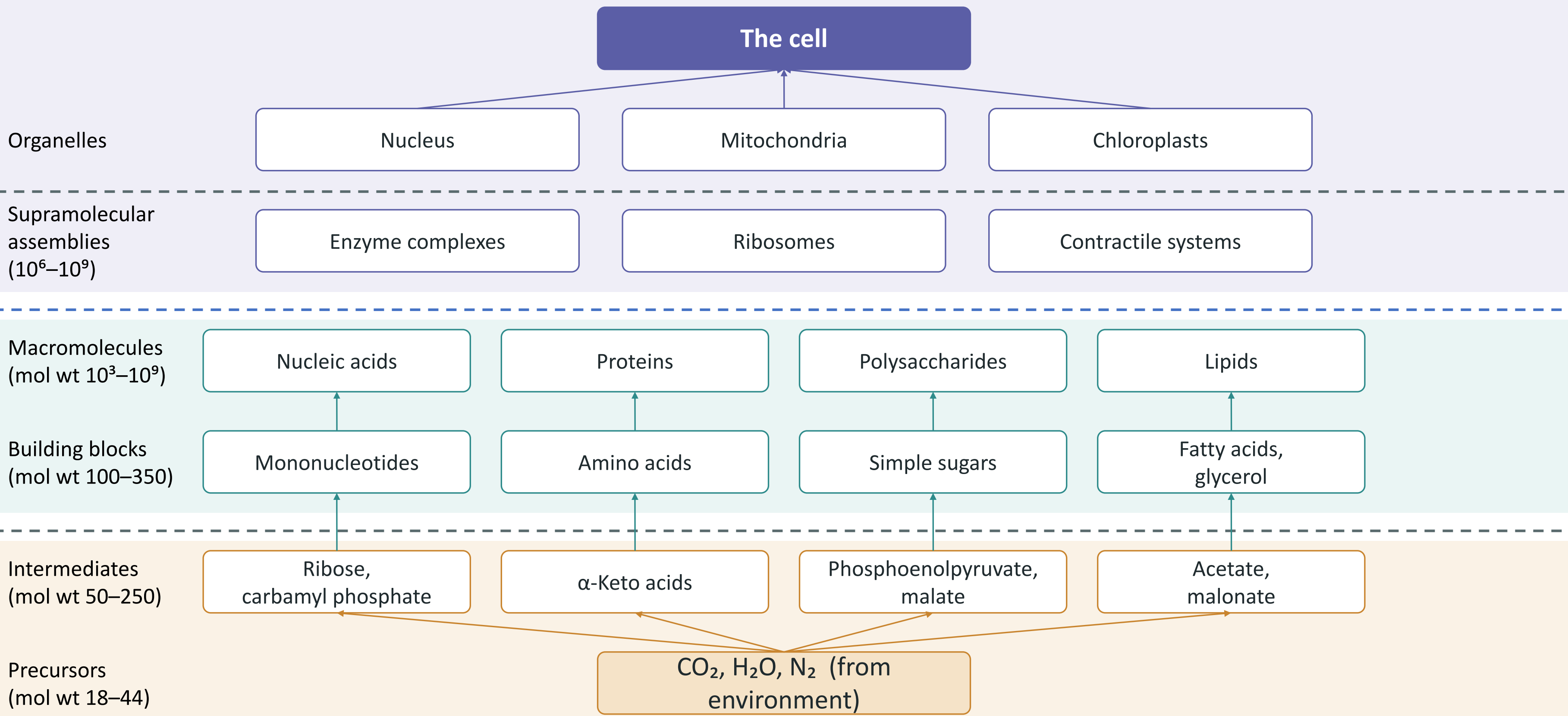
PTM = post-translational modification (e.g., glycosylation, phosphorylation, disulfide-bond formation) — largely absent in prokaryotes; this is why mammalian hosts like CHO or HEK cells are used for complex human therapeutic proteins.

Different industrially used strains of Prokaryotes and Eukaryotes

Strain / organism	Industrially important product(s)
PROKARYOTES	
<i>Escherichia coli</i>	Recombinant therapeutic proteins (e.g. insulin); standard genetic-engineering host
<i>Corynebacterium glutamicum</i>	Glutamate and other amino acids
<i>Bacillus subtilis</i>	Industrial enzymes (proteases, amylases)
<i>Streptomyces</i> spp. (<i>S. griseus</i> , <i>S. rimosus</i>)	Antibiotics (streptomycin, oxytetracycline)
EUKARYOTES — FUNGI	
<i>Saccharomyces cerevisiae</i> (yeast)	Ethanol; baking; brewing
<i>Aspergillus niger</i> (mold)	Citric acid; industrial enzymes
<i>Penicillium chrysogenum</i> (mold)	Penicillin
EUKARYOTES — OTHER	
<i>CHO cells</i> (Chinese hamster ovary)	Therapeutic proteins and monoclonal antibodies (mammalian cell culture)

From Simple Precursors to the Cell: A Hierarchy of Molecular Organization

Molecular weight and structural complexity rise together as biochemical engineers move from raw feedstock to living systems



Composition of a Rapidly Growing *E. coli* Cell

Water dominates cell mass, but proteins define what the cell actually does.

Component	% of cell weight	Avg. mol. weight	Approx. no. per cell	Distinct kinds
H ₂ O	70	18	4×10 ¹⁰	1
Inorganic ions (Na ⁺ , K ⁺ , Mg ²⁺ , Ca ²⁺ , Fe ²⁺ , Cl ⁻ ...)	1	40	2.5×10 ⁸	20
Carbohydrates & precursors	3	150	2×10 ⁸	200
Amino acids & precursors	0.4	120	3×10 ⁷	100
Nucleotides & precursors	0.4	300	1.2×10 ⁷	200
Lipids & precursors	2	750	2.5×10 ⁷	500
Other small molecules*	0.2	150	1.5×10 ⁷	200
Proteins	15	40,000	10 ⁶	2,000–3,000
DNA	1	2.5×10 ⁹	4	1
RNA (breakdown next slide)	6	—	—	—

* Heme, quinones, breakdown products of food molecules, etc. · Source: Bailey & Ollis, Table 2.12

RNA Diversity Inside the Cell

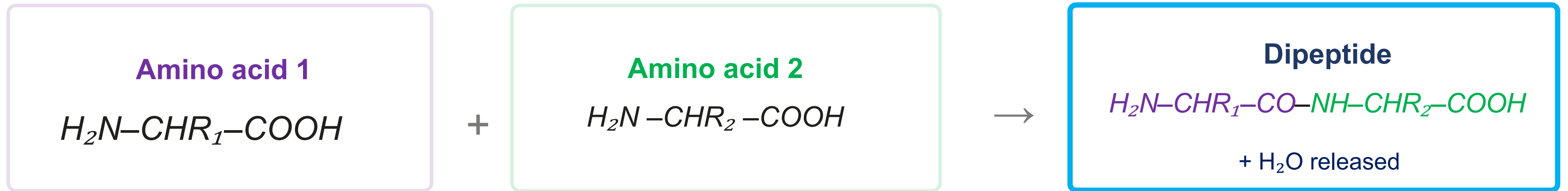
The 6% RNA share splits across four functionally distinct classes.

RNA class	Role	Avg. mol. weight	Approx. no. per cell	Distinct kinds
16S rRNA	Small ribosomal subunit scaffold	500,000	3×10^4	1
23S rRNA	Large ribosomal subunit scaffold	1,000,000	3×10^4	1
tRNA	Delivers amino acids to the ribosome	25,000	4×10^5	40
mRNA	Carries protein-coding information	1,000,000	10^3	1,000

rRNA and tRNA are structurally uniform but numerous per cell, while mRNA is the most chemically diverse — one species per gene expressed.

How Amino Acids Link: Peptide Bond Formation

A condensation reaction joins amino acids end-to-end, releasing water.

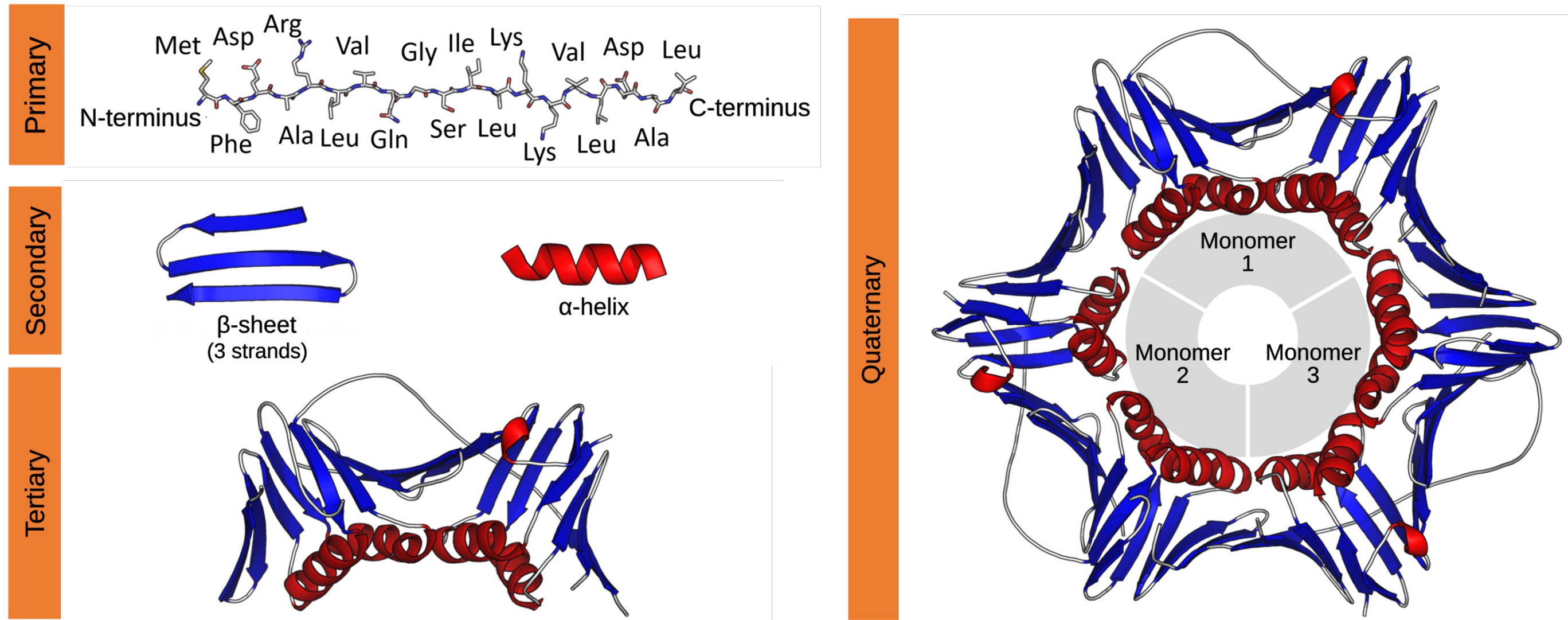


The new C(=O)–NH bond — the peptide bond — has partial double-bond character, locking the six atoms around it into a rigid plane that governs how the whole chain can fold.

Chains under ~50–100 residues are called polypeptides (e.g. insulin); longer chains are proteins.

From Sequence to Shape: Protein Structure

The amino acid sequence dictates the folded 3D shape.



Primary — the linear sequence of amino acids, N- to C-terminus.

Secondary — local folding into α -helices and β -sheets, held by backbone hydrogen bonds.

Tertiary — the complete 3D fold of one polypeptide chain.

Quaternary — multiple folded chains (monomers) assembling into one functional complex.

Fig: Types of protein structure from Primary to Quaternary structure

Antibody Architecture: The Immunoglobulin

Two heavy chains + two light chains, held together by disulfide bonds.

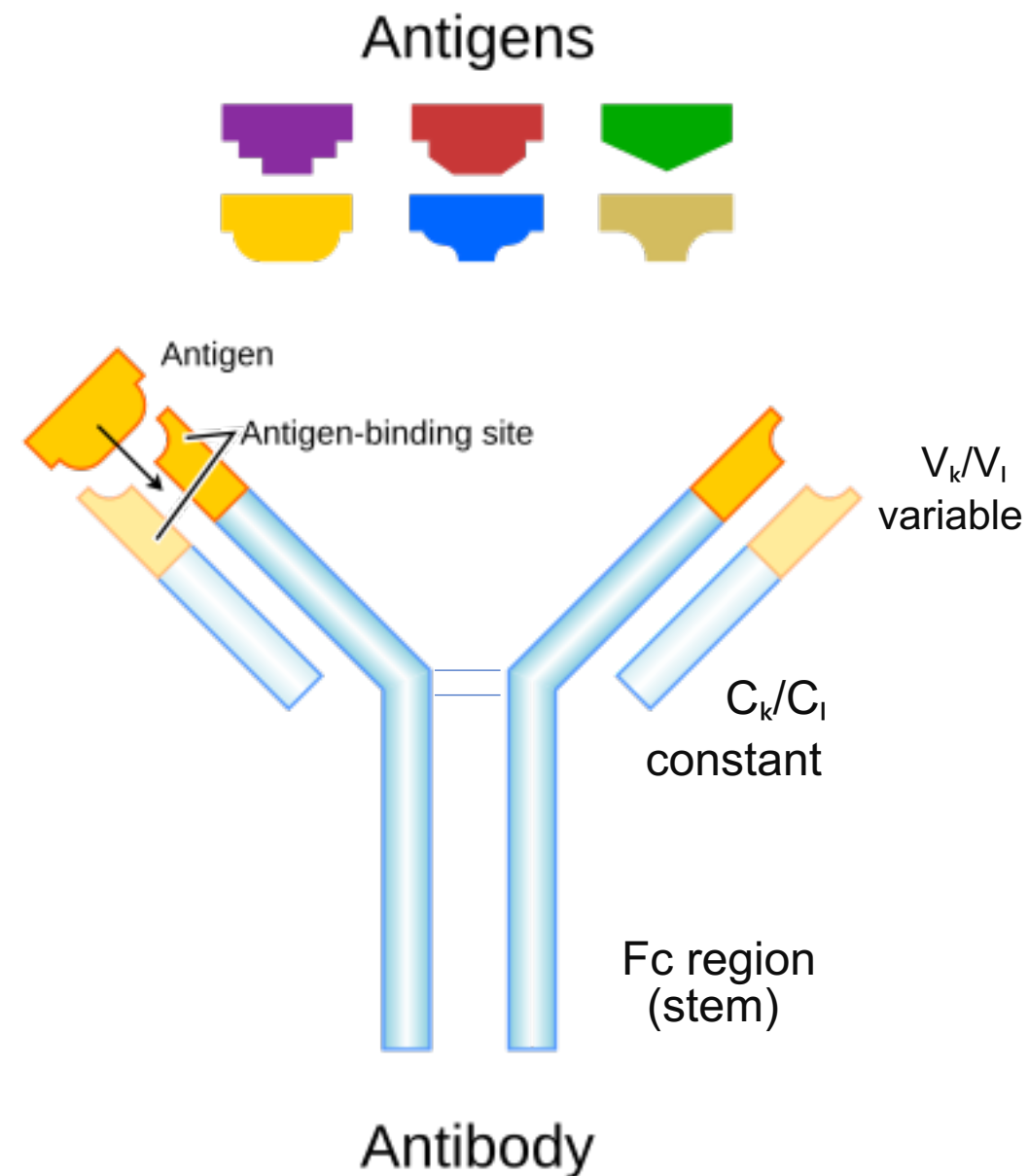


Fig: 2D schematic diagram of an Antibody

The variable (V) regions at the tips of the chains form the antigen-binding site, which is where each antibody's specificity is derived.

C (constant) regions and the Fc stem are the same across antibodies of one class.

The Antibody in Three Dimensions

IgG folds into three compact domains connected by a flexible hinge.

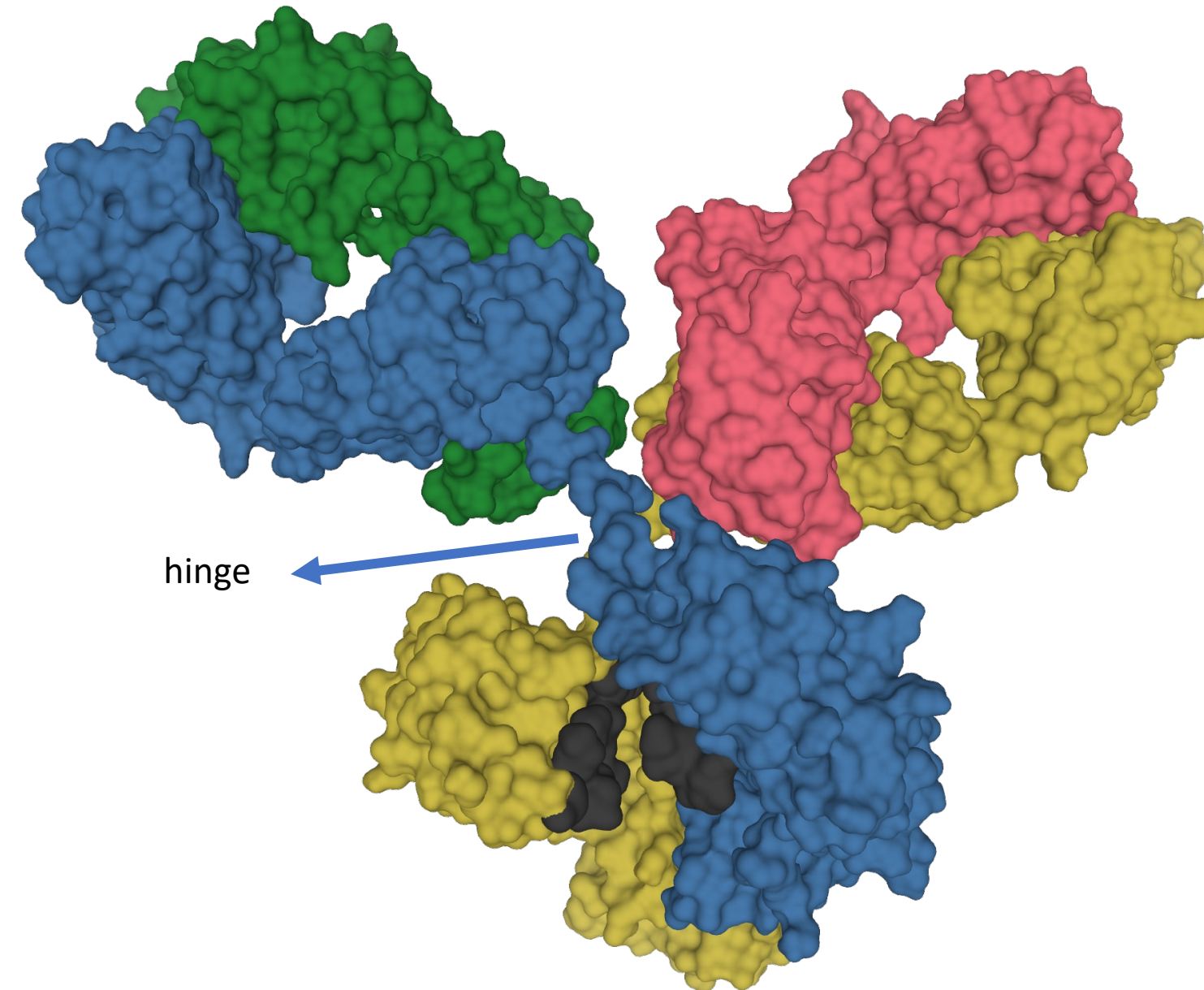


Fig: 3D diagram of an Antibody

The hinge allows the two antigen-binding arms to swing independently, so one antibody can bind two separate antigen sites at once

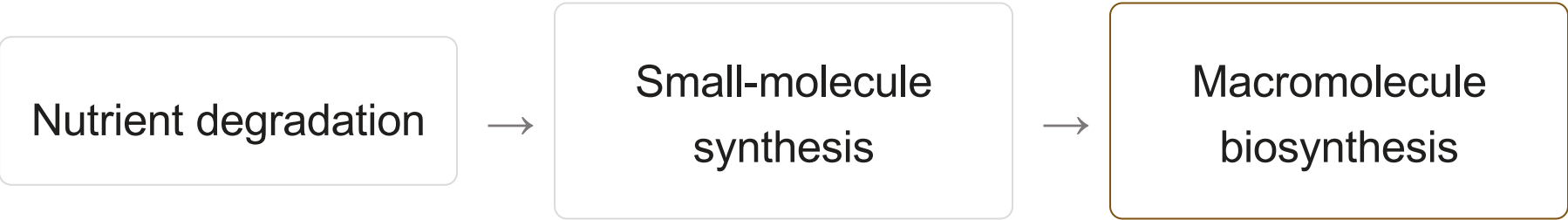
What Cells Need: Nutrients and the Metabolic Roadmap

Four elements dominate cell nutrition; every cell routes them through the same three-stage roadmap.

Table 5.1 — Typical elemental sources

Element	Typical source
Carbon	CO ₂ , sugars, proteins, fats
Nitrogen	Proteins, NH ₄ ⁺ , NO ₃ ⁻
Sulfur	Proteins, SO ₄ ²⁻
Phosphorus	PO ₄ ³⁻

Metabolic roadmap (Fig. 5.2)



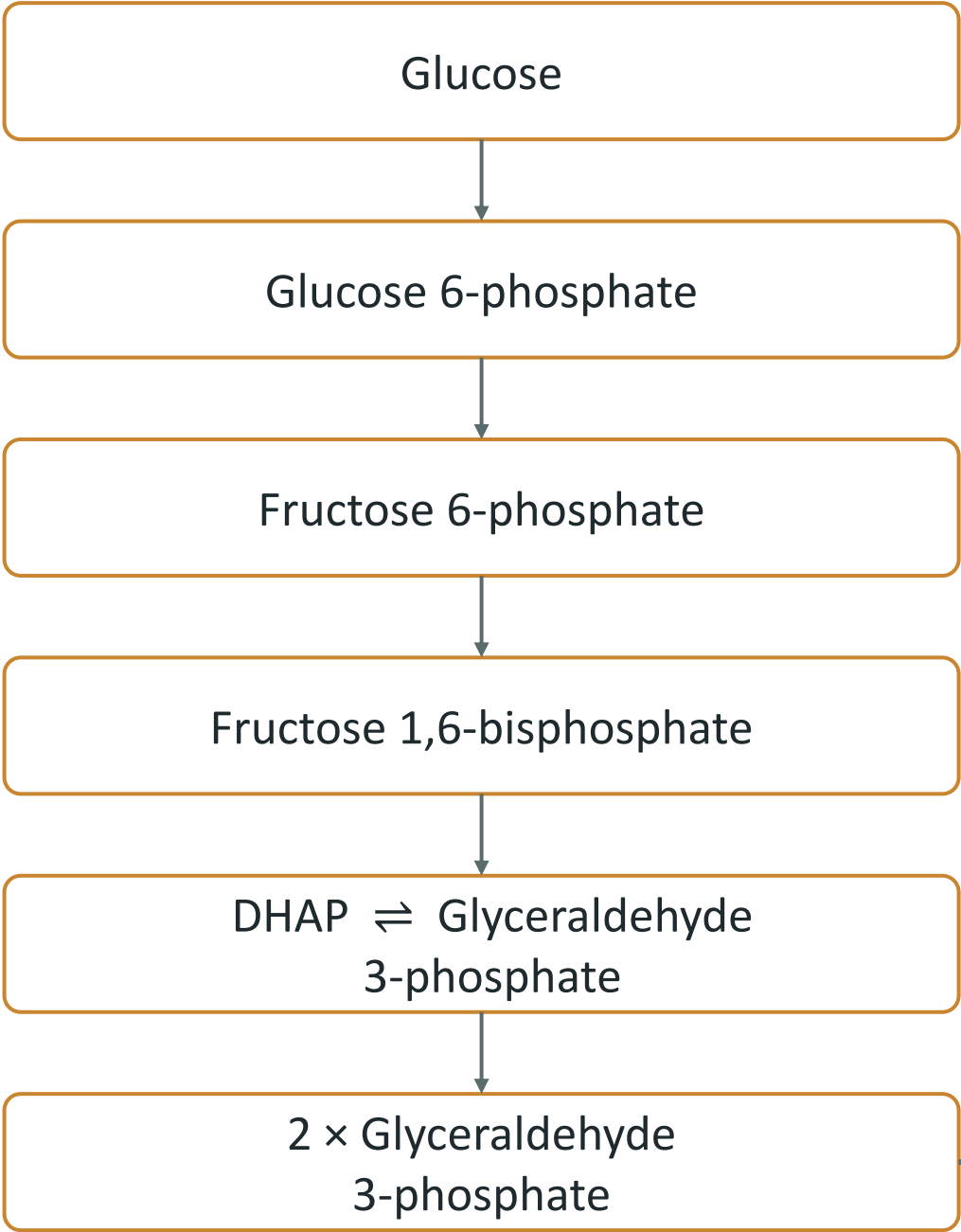
~1,000+ distinct enzyme-catalyzed reactions carry out this roadmap in a typical cell.

Heterotrophs get carbon from organic compounds; autotrophs need only CO₂. Combined with energy source (chemical vs. light), this defines a cell's full metabolic classification (Table 5.2).

Glycolysis: The Embden-Meyerhof-Parnas (EMP) Pathway

One six-carbon glucose splits into two three-carbon units, each of which runs the full energy-payoff sequence

ENERGY INVESTMENT PHASE



Hexokinase
ATP $\rightarrow$ ADP

Phosphohexose isomerase

Phosphofructokinase
ATP $\rightarrow$ ADP

Aldolase (splits chain)

Triose phosphate isomerase

**$\times 2$
per
glucose**

G3P dehydrogenase
NAD⁺ $\rightarrow$ NADH

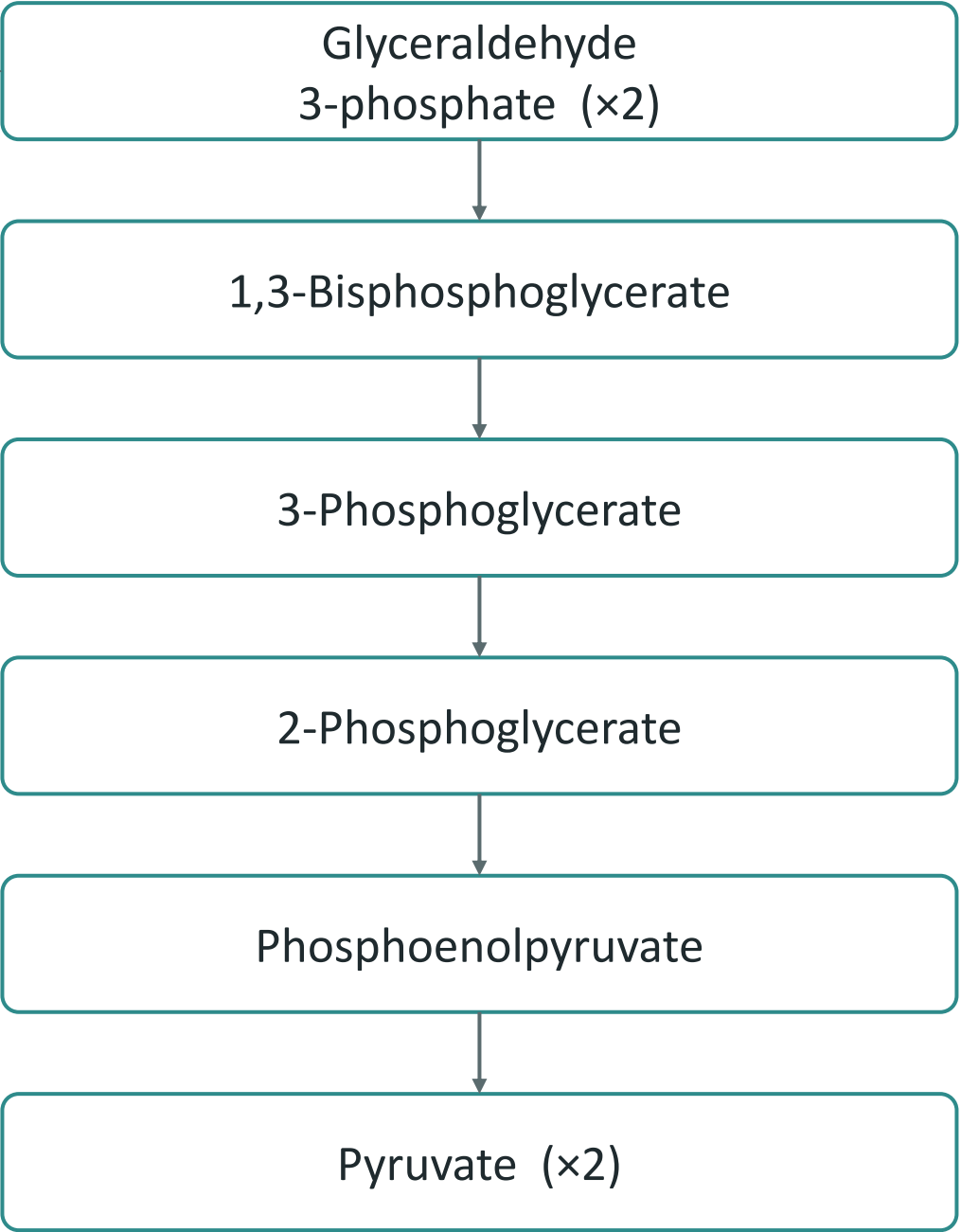
Phosphoglycerate kinase
ADP $\rightarrow$ ATP

Phosphoglycerate mutase

Enolase
- H₂O

Pyruvate kinase
ADP $\rightarrow$ ATP

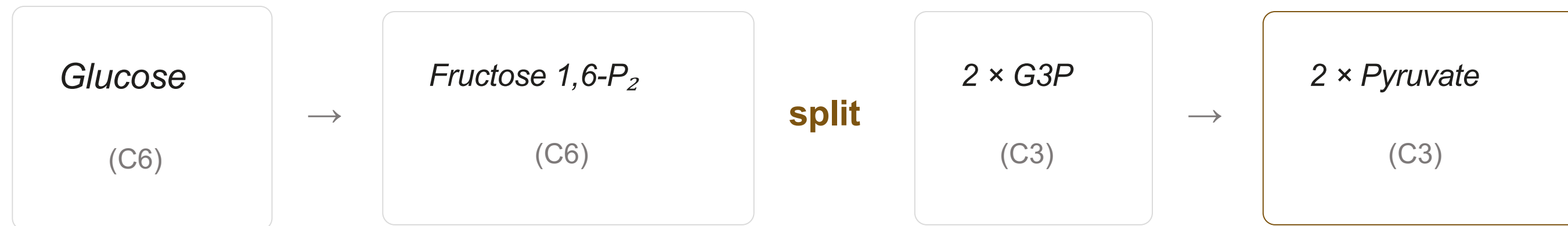
ENERGY PAYOFF PHASE ($\times 2$)



Net yield per glucose: 2 ATP consumed (hexokinase, PFK) vs. 4 ATP produced (phosphoglycerate kinase $\times 2$, pyruvate kinase $\times 2$) = **net 2 ATP, 2 NADH, and 2 pyruvate.**

Glycolysis: Harvesting Energy from Glucose

The Embden-Meyerhof-Parnas (EMP) pathway splits glucose into two 3-carbon units.



Net per glucose: 2 ATP (substrate-level phosphorylation) + 2 NADH.
About 31% of the free energy from glucose breakdown is conserved this way (~53% after correcting for real in-cell concentrations).

Glycolytic intermediates and pathway steps are also biosynthetic starting materials — this dual role is why EMP is called "amphibolic."

Spending ATP to Build Molecules: Biosynthesis

Example: the aspartate family pathway builds 5 amino acids, each step paid for in ATP.

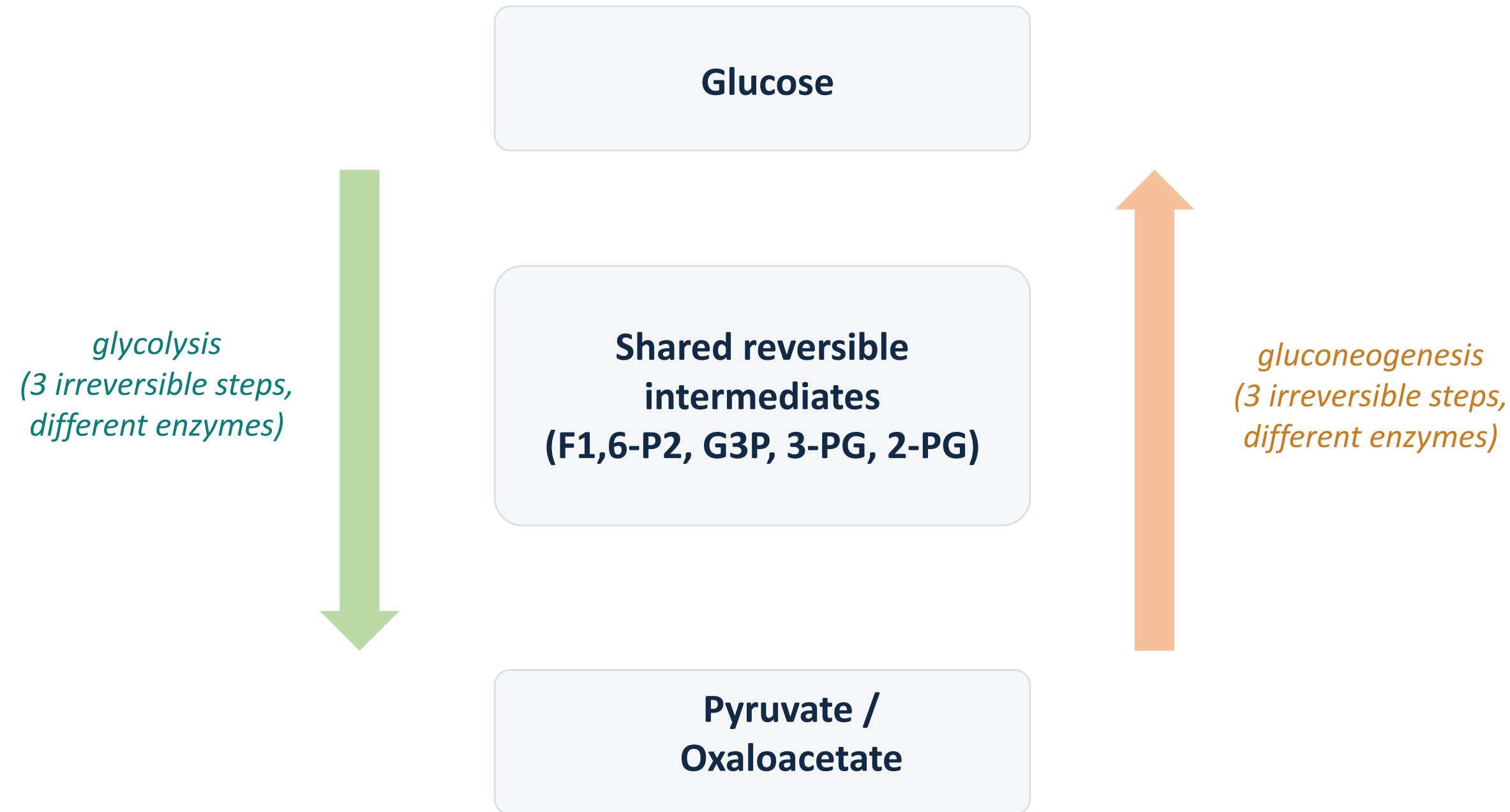


Each arrow in the real pathway (Fig. 5.12) represents an enzyme-catalyzed step, and several consume one ATP to "activate" an intermediate — converting a chemically unreactive group into one that reacts readily.

General principle: biosynthesis is energy-expensive — typically 1–2 high-energy phosphate bonds hydrolyzed per bond formed.

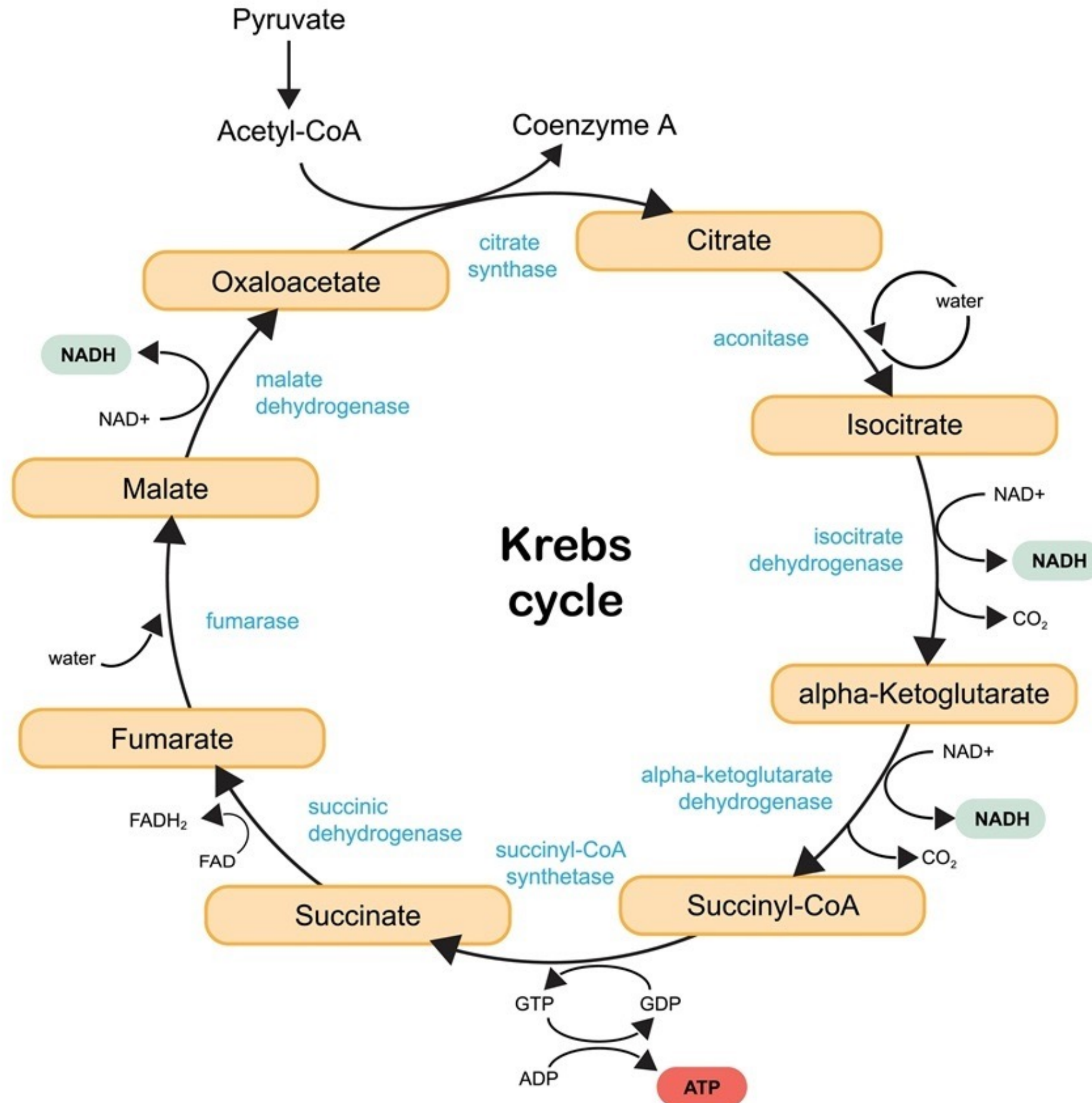
Two-Way Street: Glycolysis and Gluconeogenesis

Breaking down glucose and building it back up share most of the same steps run in reverse.

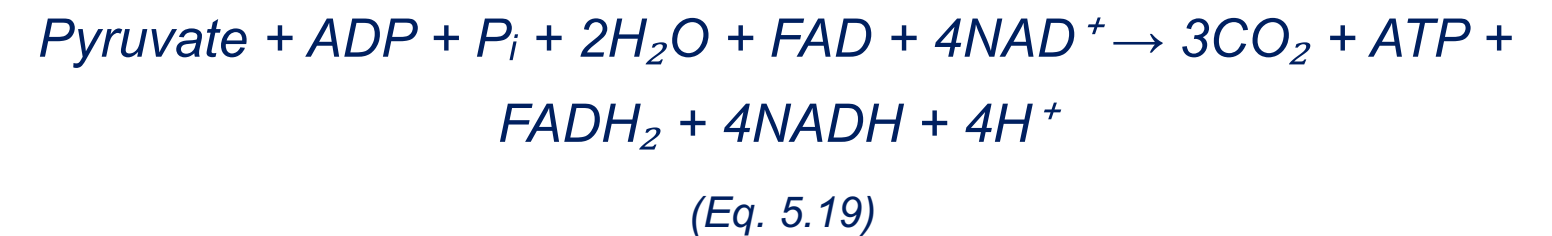


The near-equilibrium steps are shared and freely reversible. The 3 far-from-equilibrium steps in each direction use different, independently regulated (often allosteric) enzymes — this lets a cell run synthesis and breakdown at different rates simultaneously.

The TCA (Krebs) Cycle



Pyruvate's remaining carbons are fully oxidized to CO₂, harvesting electrons at every turn.

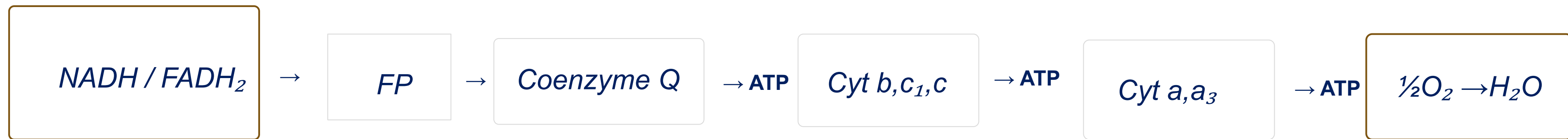


Per turn (per pyruvate): 3 CO₂ released, 1 ATP (substrate-level), 4 NADH, 1 FADH₂ — all destined for the respiratory chain.

Since glycolysis produces 2 pyruvate per glucose, double these numbers for each glucose oxidized.

Oxidative Phosphorylation:

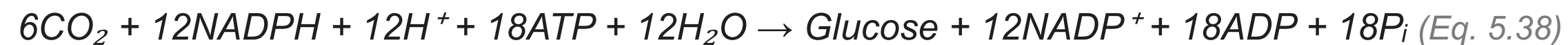
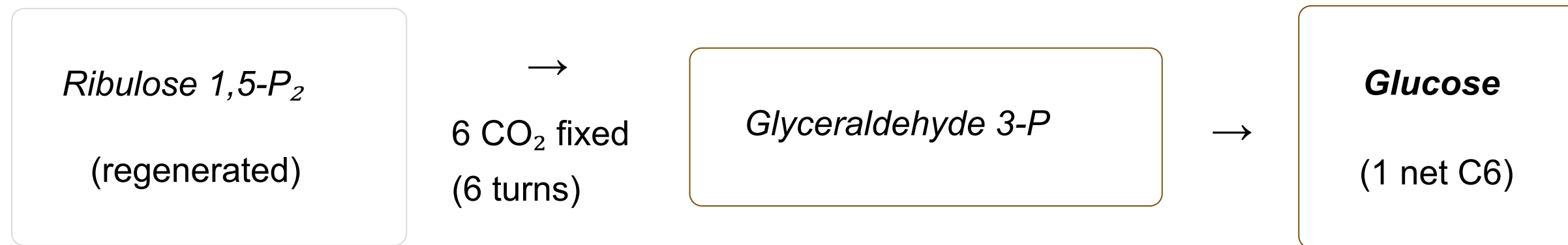
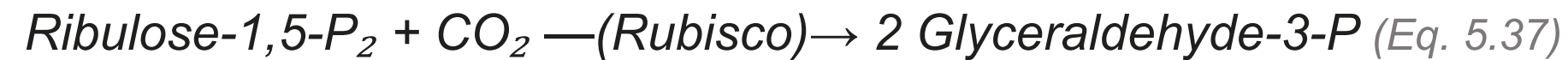
NADH and FADH₂ hand off electrons through a carrier chain, paying for ATP along the way.



Full oxidation of glucose (EMP + TCA + respiratory chain): $\text{Glucose} + 38 \text{ ADP} + 38 \text{ P}_i \rightarrow 6 \text{ CO}_2 + 38 \text{ ATP} + 44 \text{ H}_2\text{O}$ (Eq. 5.23) — about 19× more ATP than glycolysis alone, at ~40% energy-capture efficiency (~70% corrected).

Running Respiration in Reverse: The Calvin Cycle

CO₂ fixation for glucose synthesis — an investment of ATP and NADPH, not a source of it.



Contrast with TCA: the Calvin cycle spends 18 ATP + 12 NADPH per glucose built, rather than generating them — the mirror image of respiration's energy economics.

Bioenergetics Scoreboard: Glycolysis vs. Full Respiration

Same starting molecule, very different energy payout depending on whether O₂ is available.

Pathway	Net reaction (per glucose)	ATP yield	Efficiency
Glycolysis only (anaerobic)	$Glucose \rightarrow 2\ Pyruvate + 2\ ATP + 2\ NADH$	2 ATP substrate-level	~31% (~53% corrected)
Full respiration (EMP+TCA+ETC)	$Glucose + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$	38 ATP mostly oxidative	~40% (~70% corrected)

Respiration extracts roughly 19× more ATP from the same glucose molecule than fermentation alone — the respiratory chain lets the cell fully oxidize glucose to CO₂ instead of stopping at pyruvate/lactate.

Bioenergetics Scoreboard: TCA vs. Calvin Cycle

One cycle harvests energy from carbon; the other spends energy to build it.

Cycle	Direction	Final reaction	Net energy carriers
TCA (Krebs)	Catabolic harvests energy	$Pyruvate + ADP + P_i + 2H_2O + FAD + 4NAD^+ \rightarrow 3CO_2 + ATP + FADH_2 + 4NADH + 4H^+$	+1 ATP, +4 NADH, +1 FADH ₂ per turn
Calvin	Anabolic invests energy	$6CO_2 + 12NADPH + 12H^+ + 18ATP + 12H_2O \rightarrow Glucose + 12NADP^+ + 18ADP + 18P_i$	-18 ATP, -12 NADPH per glucose

The sign convention matters for teaching: TCA's carriers are a yield (+), Calvin's are a cost (−) — together they show that "cycle" describes the chemistry, not the energy direction.

Real Cells Aren't at Standard Conditions

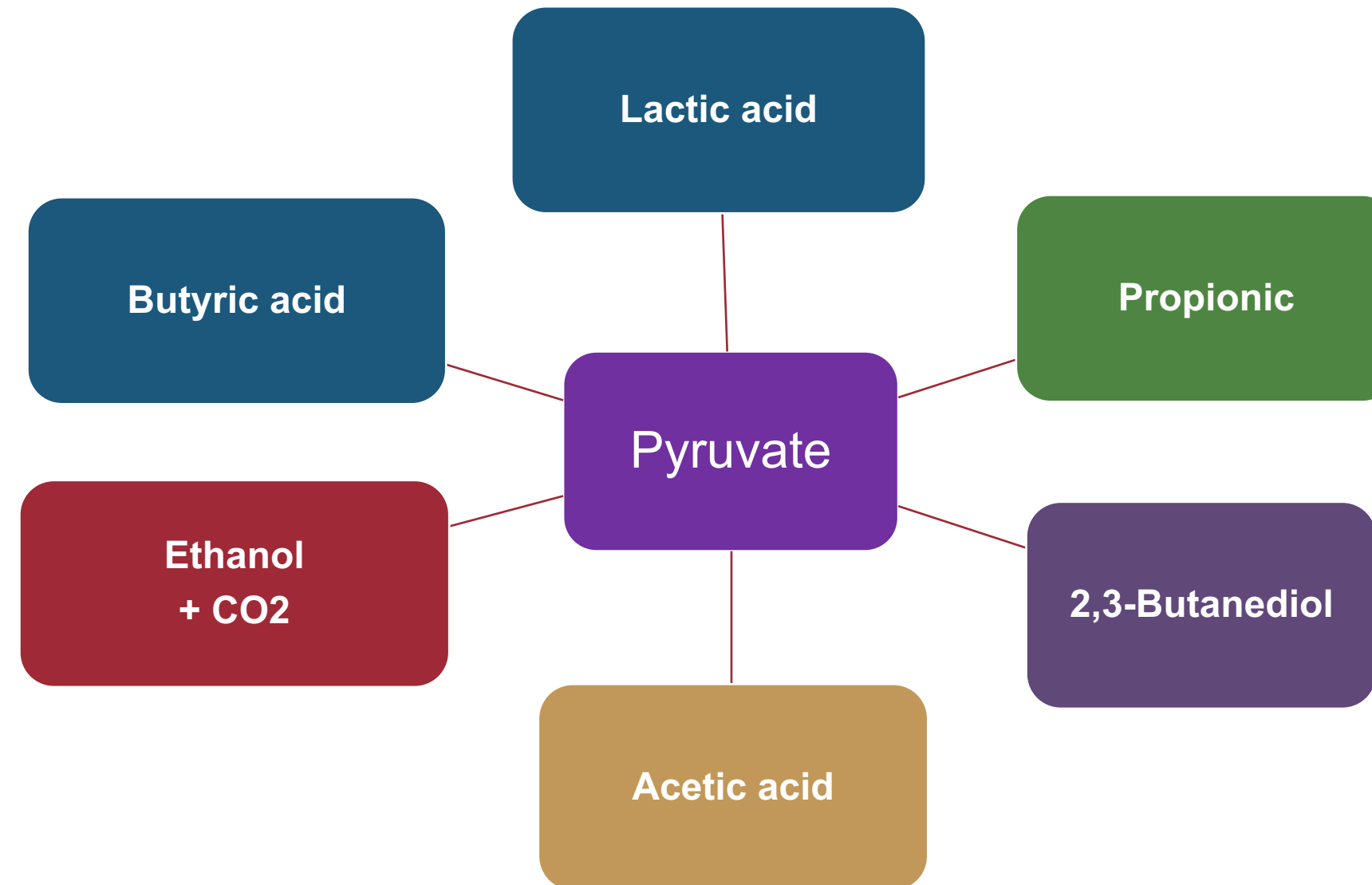
Why glycolysis is more efficient in vivo than textbook ΔG° values suggest.

Intermediate	Concentration (μM)	ΔG° for step
Glucose	3,830	— non-equilibrium
Glucose 6-P	830	near 0 (reversible)
Fructose 6-P	140	near 0 (reversible)
Fructose 1,6-P ₂	31	— non-equilibrium
3-Phosphoglycerate	118	near 0 (reversible)
Phosphoenolpyruvate (PEP)	23	— non-equilibrium

Most glycolytic steps are near equilibrium in real cells: only glucose→G6P, F6P→FBP, and PEP→pyruvate are strongly non-equilibrium. That's why in-cell efficiency (~53%) beats the naive standard-state estimate (~31%).

From Pyruvate: Fermentation End Products

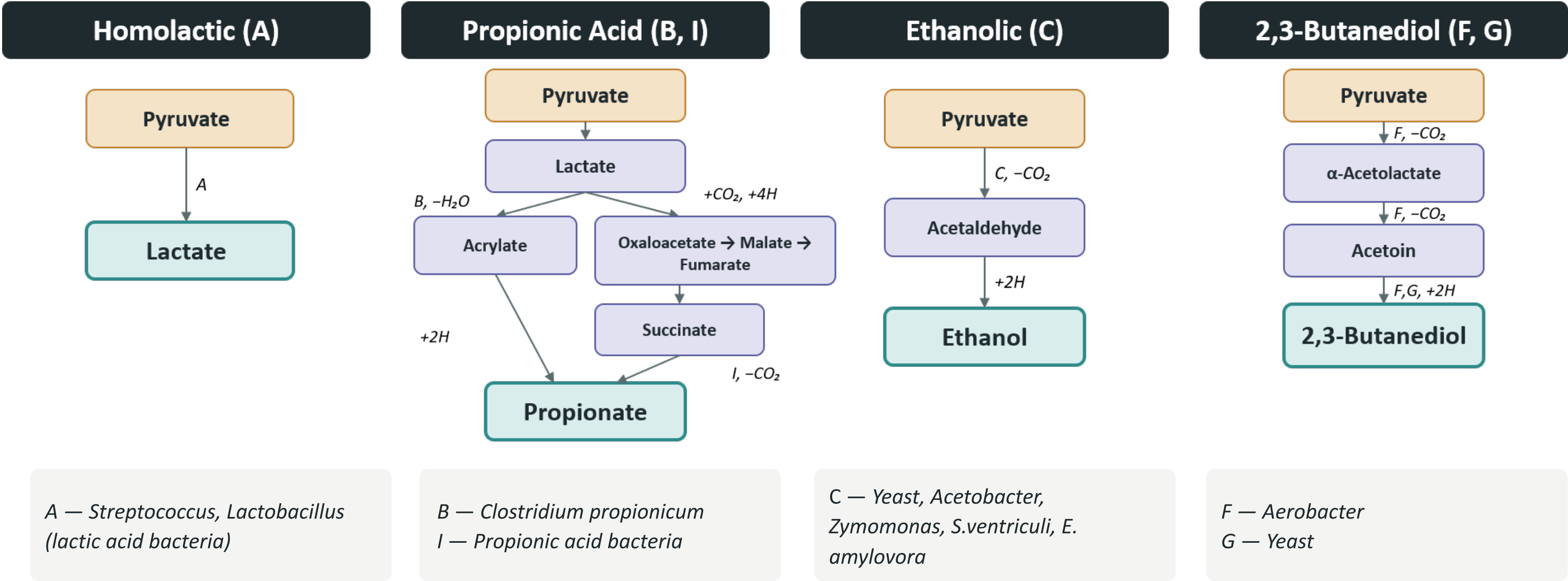
One branch point, many possible fates — the organism determines which way electrons flow.



Circled end products (schematic) all regenerate NAD^+ from the NADH produced during glycolysis — without this, glycolysis would stall for lack of oxidized NAD^+ .

Products depend on both organism and conditions — e.g., yeast shifts from ~172 to ~150 mmol ethanol (per 100 mmol glucose) as pH rises from 3.0 to 7.0, with glycerol yield rising in exchange.

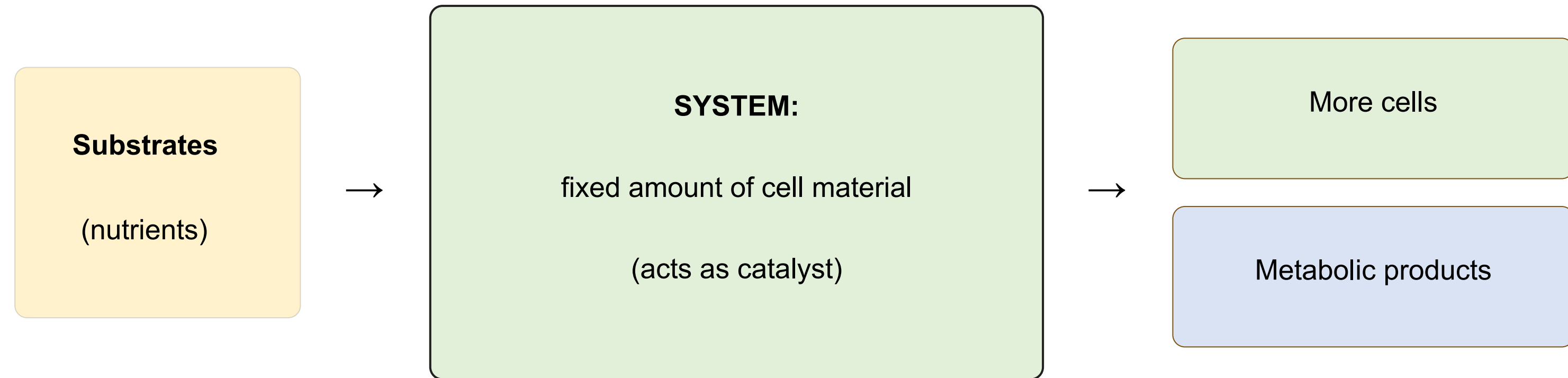
Bacterial sugar fermentations which proceed through the EMP pathway



Different end products, same purpose: every route reoxidizes the NADH generated in glycolysis, so fermentation can continue without O₂.

Zooming Out: Cell Growth as a Black Box

For stoichiometry, cell material itself can be treated as the catalyst of its own growth.



This "black box" view underlies stoichiometric medium design: to grow a target mass of cells (X) with elemental weight fraction w_i , at minimum $w_i(X)$ of element i must be supplied in the nutrients.

Where Does the Substrate Actually Go?

Not all consumed substrate is used to build new cells; some is used for energy and maintenance.

$$(\Delta S) = (\Delta S)_{\text{assimilation}} + (\Delta S)_{\text{growth energy}} + (\Delta S)_{\text{maintenance energy}} \text{ (Eq. 5.47)}$$

$$1/Y_{X/S} = (\Delta S)_{\text{assim}}/(\Delta X) + (\Delta S)_{\text{growth}}/(\Delta X) + (\Delta S)_{\text{maint}}/(\Delta X) \text{ (Eq. 5.48)}$$

Symbol	Meaning
ΔS	total substrate consumed
ΔX	cell mass produced
$Y_{X/S}$	observed growth yield factor, $\Delta X/\Delta S$
Assimilation/assim	substrate carbon actually built into new cell mass
growth energy	substrate burned for the energy of making new cells
maintenance energy/maint	substrate burned just to stay alive (ion pumping, repair) — not tied to growth

A Full Elemental Balance for Growth and Product

The most rigorous view: balance every C, H, O, and N atom across substrate, cells, and product.

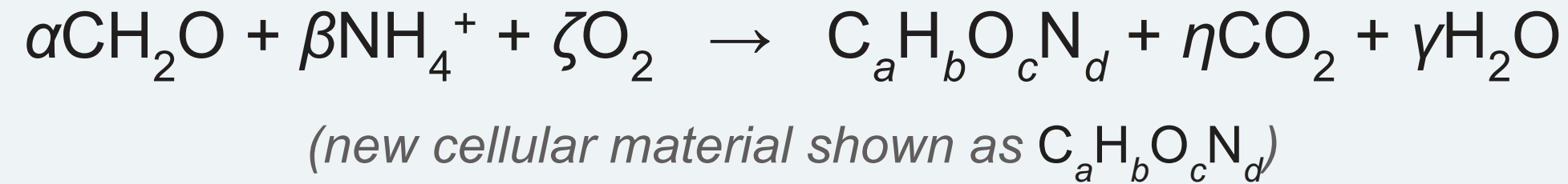


Symbol	Meaning
$a'-f'$	stoichiometric coefficients, fixed by balancing C, H, O, N and available electrons
$CH_a O_b$	substrate, per mole of carbon (e.g. glucose as CH ₂ O)
$H_c O_d N_e$	nitrogen source (e.g. NH ₃ , NO ₃ ⁻)
$CH_\alpha O_\beta N_\gamma$	empirical formula of dry cell mass (biomass), per mole of carbon
$CH_\gamma O_\delta$	empirical formula of the excreted product, per mole of carbon

Writing every formula per mole of carbon keeps the balance dimensionally consistent regardless of molecule size.

Mass Balances: Stoichiometry

The major elemental demands of a growing cell are carbon, oxygen, hydrogen, and nitrogen. Assuming the carbon source to be characterized by (CH_2O) and nitrogen by NH_4^+ , growth follows the general balance below.



(a) Write down the restriction among the coefficients α , β , ... from consideration of elemental conservation. Given the “composition” of a cell, what additional information is needed to solve for the coefficient values?

(b) Repeat part (a) for photosynthetic systems:

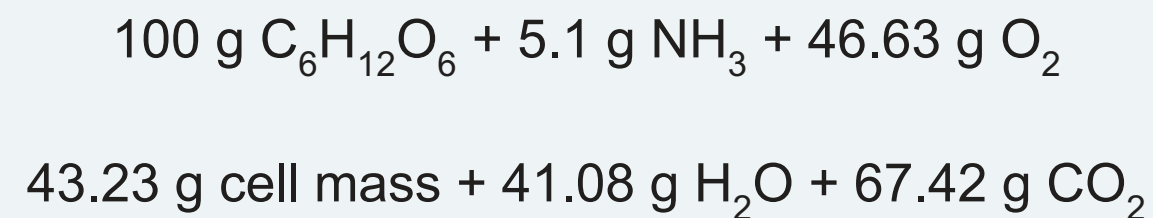


(c) Describe an explicit series of experiments by which you would evaluate the stoichiometric formula of the cell (a, b, c, d).

Stoichiometry: Aerobic vs. Anaerobic Species

To conveniently calculate nutrient consumption, aeration, and heat-transfer rates, a general stoichiometric equation can be formulated by assigning the cell an apparent molecular formula.

(a) Calculate the cell formula for growth of yeast on sugar, given the following balance (1 mol of cell produced):



(b) If the above reaction produces 0.30017 “mol” of cell, what cell formula results?

Note: elemental (C, H, O, N) and mass balances on the part-(a) reaction give the mole quantities of each species; dividing by 0.30017 mol cell normalizes the empirical formula $\text{C}_a\text{H}_b\text{O}_c\text{N}_d$ to one mole of cell.

(c) Anaerobic fermentations also yield partially oxygenated by-products. Keeping the cell formula from (b), find the unknown coefficients (mole quantities) below:

0.55556 glucose + α ammonia

→ β cell mass + 0.05697 glycerol

+ γ butanol (alcoholic product)

+ ε succinic acid (acid product)

+ 0.01164 water + 1.03076 carbon dioxide + 1.00380 ethanol

Coefficients α , β , γ , and ε are the unknowns to be solved by elemental balance

Network Mass Balance

You have a friend who asked to run a fermentation in a laboratory course but neglected to weigh the glucose he added and forgot to analyze for ethanol.

He has found that the fermentation produced the compounds shown in Table. He knows that all glucose ($C_6H_{12}O_6$) is oxidized to pyruvic acid ($CH_3COCOOH$) by the EMP pathway, and that no other products except ethanol are formed.

He comes to you for help in salvaging something out of this mess. **How many moles of ethanol should you tell him have been formed?** *(State your assumptions.)*

TABLE

Compound	Moles
Lactic acid ($CH_3CHOHCOOH$)	10
Acetic acid (CH_3COOH)	5
Carbon dioxide (CO_2)	15
Hydrogen (H_2)	10

All quantities in the table are given in moles. Ethanol was not measured, and glucose consumed was not weighed — both must be recovered from the EMP stoichiometry and an elemental (C, H, O) balance over the network.

